

Arithmetic and complexity

An explicit rational Valiant-rigid matrix family

Difficulty: extreme **Importance:** broadly interesting**Status:** Open

Rating rationale: Extreme because an explicit family at Valiant’s rank scale would break a longstanding linear-circuit lower-bound barrier; broad importance connects sparse perturbations, matrix structure and computational complexity.

Provenance: rational-field specialization of a source-stated explicit-rigidity target.

For $A \in \mathbb{Q}^{n \times n}$ and an integer $0 \leq r \leq n$, define its rigidity over $\mathbb{Q}$ by

$$R_A^{\mathbb{Q}}(r) = \min \left\{ |\{(i, j) : A_{ij} \neq B_{ij}\}| : B \in \mathbb{Q}^{n \times n}, \text{rank}_{\mathbb{Q}}(B) \leq r \right\}.$$

Construct a constant $\delta > 0$ and a family $A_n \in \mathbb{Q}^{n \times n}$ such that a single deterministic algorithm, on input the unary encoding 1^n , outputs all rational entries of A_n in binary using polynomially many bit operations in n , and

$$R_{A_n}^{\mathbb{Q}} \left(\left\lfloor \frac{n}{\log \log n} \right\rfloor \right) \geq n^{1+\delta}$$

for every sufficiently large integer n . Here $\log$ denotes the natural logarithm. The output convention requires polynomial total bit length; an exponentially long algebraic description does not meet this requirement.

Rigidity measures how resistant a matrix is to rank reduction by sparse entry changes. An explicit family at this scale would supply hard instances for linear circuit computation. The source states the target over a field; this entry explicitly selects rational entries and rational perturbations, rather than asserting that all choices of field are equivalent.

References

1. J. Alman and J. Liang, *Low Rank Matrix Rigidity: Tight Lower Bounds and Hardness Amplification*, arXiv:2502.19580 (2025), Section 1 and Section 1.1: definition of rigidity, uniform polynomial-time explicitness, and the $n/\log \log n$ Valiant target. [Paper](#).
2. L. Hambardzumyan, K. Myasnikov, A. Riazanov, M. Shirley, and A. Shraibman, *Spiky Rank and Its Applications to Rigidity and Circuits*, ICALP 2026, LIPIcs 374, article 106, Section 1.2.1: explicit rigidity remains insufficient for the relevant circuit lower bounds. [Published paper](#).
3. L. G. Valiant, *Graph-theoretic arguments in low-level complexity*, MFCS 1977, pp. 162–176, the original rigidity connection to linear circuits. [Article](#).
4. Z. Dvir and A. Liu, *Fourier and Circulant Matrices are Not Rigid*, arXiv:1902.07334 (2019), main non-rigidity results for prominent structured candidates. [Paper](#).

Status check — 2026-09-10

Rechecked [Alman–Liang, §1](#) and the [ICALP 2026 spiky-rank paper](#), and searched for explicit rational Valiant-rigid constructions. The former’s low-rank bounds and conditional amplification do not meet the displayed $n/\log\log n$ rank scale. No polynomial-time rational family meeting the target was located. The conclusion concerns rigidity over the rationals; results in other fields or for random matrices require a separate comparison.